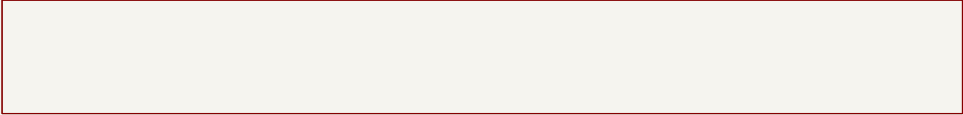


MACH3-SIMPLE – 6 axes, two LPT ports, buffered 5 V outputs

A copy of the board in service, extras removed. NOT ISOLATED on the output side.

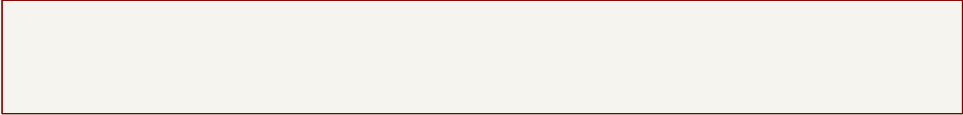
A1: NOT BUILT, NOT TESTED. 24 V in only; isolated VFD port.

Power – 24 V in, 5 V made on board



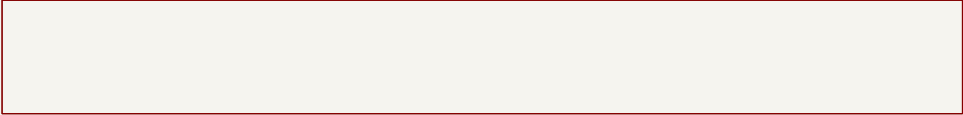
File: 01_power.kicad_sch

Inputs – home, E-stop, THC, handwheel



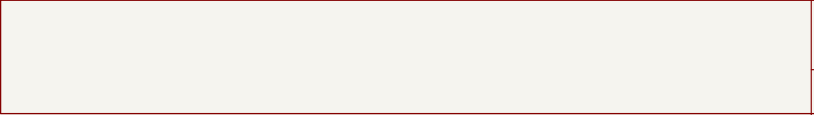
File: 03_inputs.kicad_sch

Spindle – isolated 0–10 V and FWD–REV for the VFD



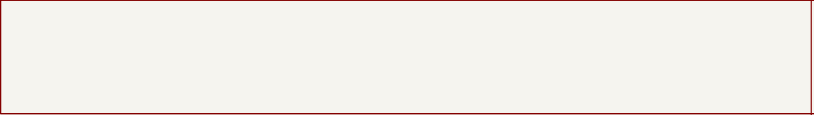
File: 05_spindle.kicad_sch

LPT port 1 – 6 axes, buffered 5 V out

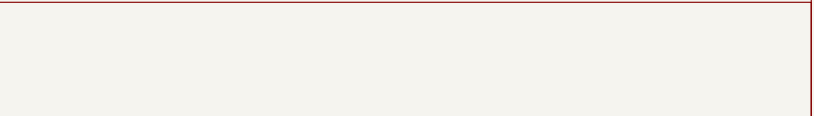


File: 02_outputs.kicad_sch

RS232, MCU and six relays



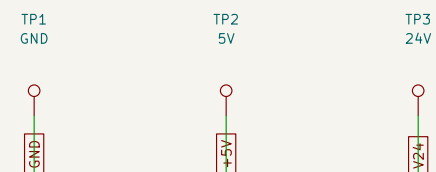
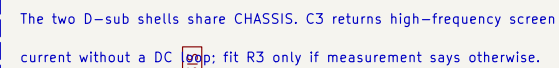
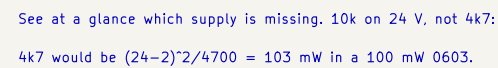
File: 04_relays.kicad_sch



Sheet: /		
File: MACH3SIMPLE.kicad_sch		
Title: MACH3-SIMPLE: copy of the board in service		
Size: A4	Date:	Rev: A1 ENGINEERING ONLY
KICad E.D.A. 10.0.6		Id: 1/6

One common ground: the outputs are not isolated, so there is nothing to keep apart. See the header of `build_project.py` for what that costs.

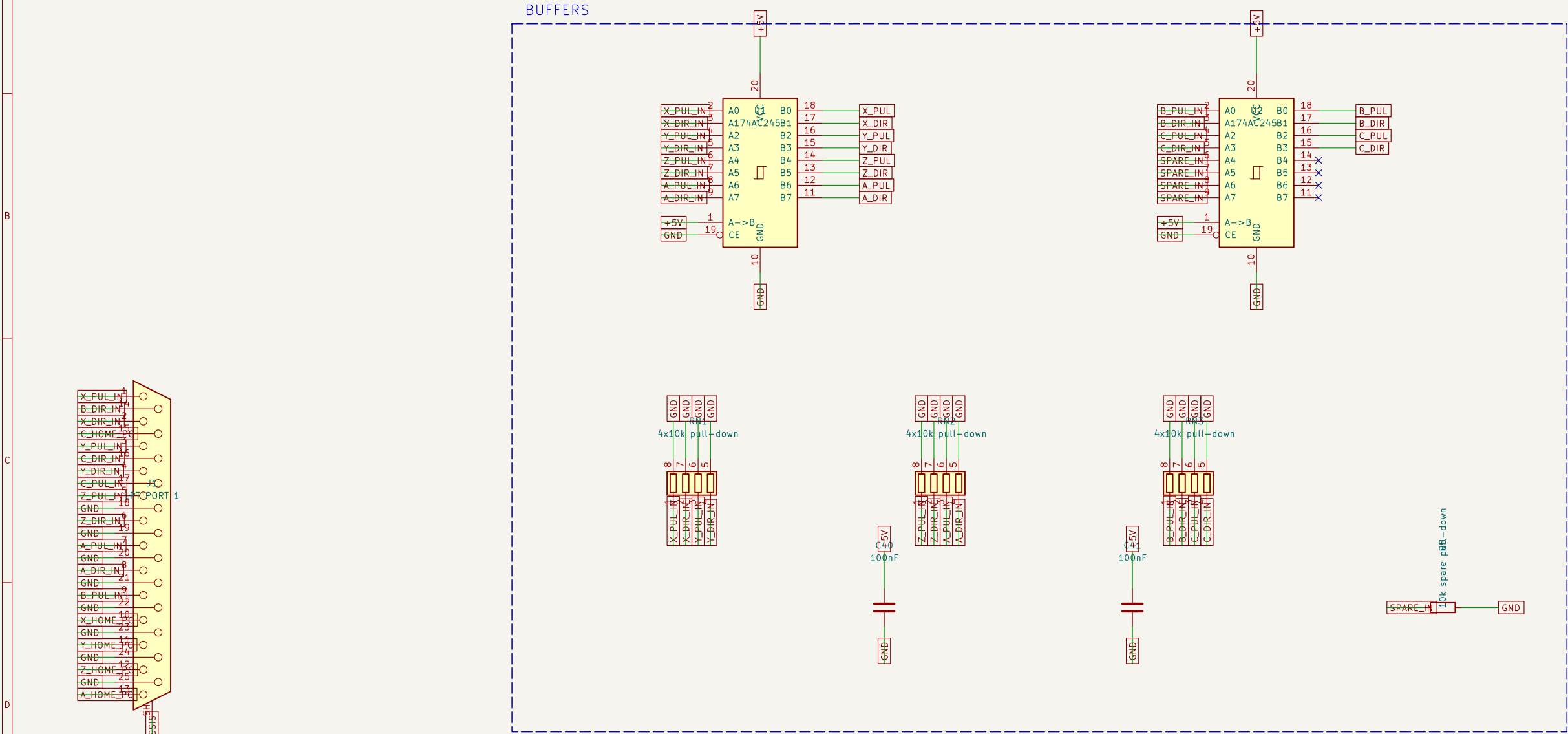
Feeds the buffers, the optocoupler LEDs on the PC side, the MCU, the RS232 transceiver and the handwheel. An LM2596S-5.0 buck off V24, so the board takes one supply. About 0.4 W lost at 0.4 A, where a 7805 would burn 7.6 W.



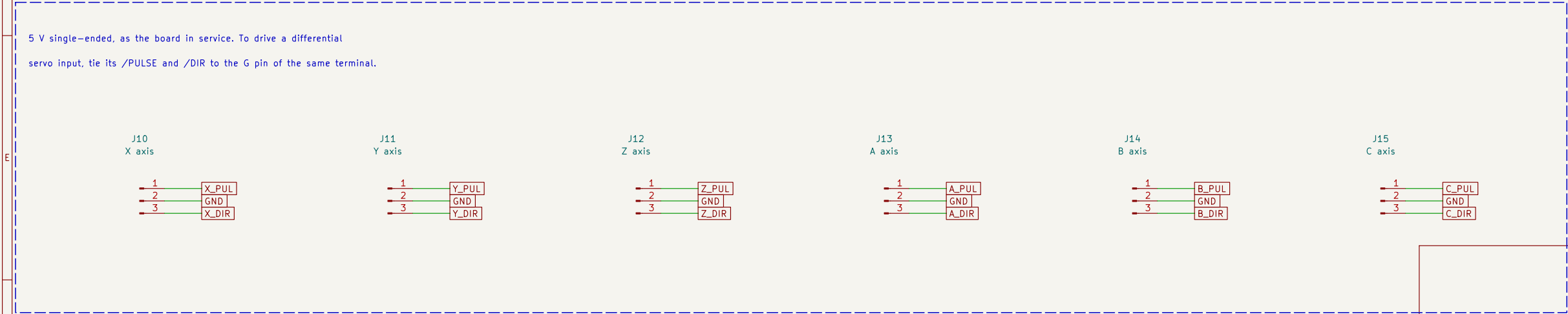
LPT PORT 1 – MOTION. Pin numbers are the ones printed on the board in

service, so the existing Mach3 profile carries over unchanged.

NOT ISOLATED: the axis terminals share ground with the PC. That is the arrangement on the board in service and the one chosen for this copy.



AXIS TERMINALS – PULSE / GND / DIR



INPUTS. 24 V sensors on optocouplers, handwheel straight through.

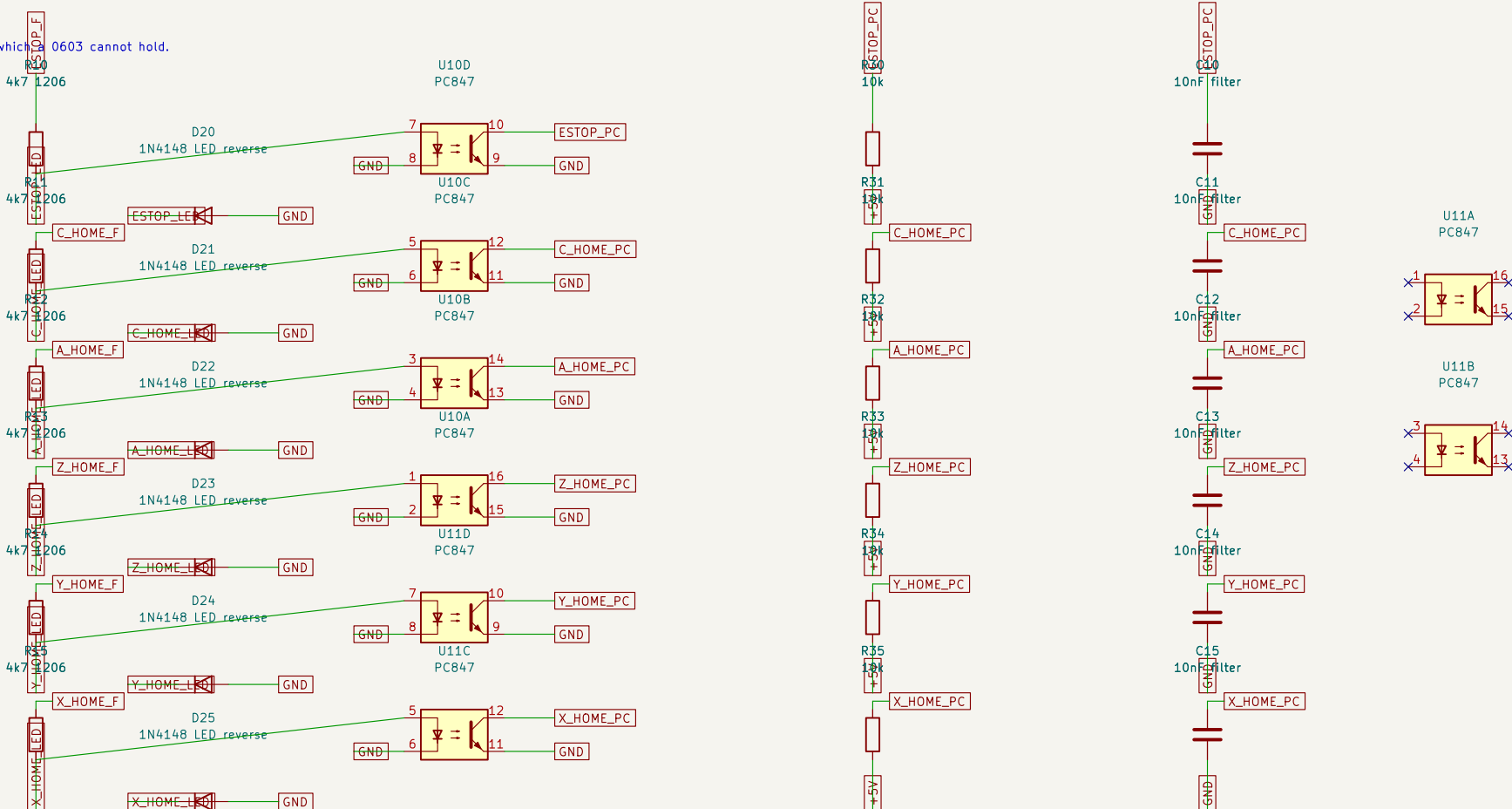
EVERY optocoupler LED gets an anti-parallel diode. A PC847 LED is rated

6 V reverse and a sensor cable beside a motor lead delivers far more.

That is the most likely reason optocouplers keep dying, and the fix is

24 V INPUT CHANNELS

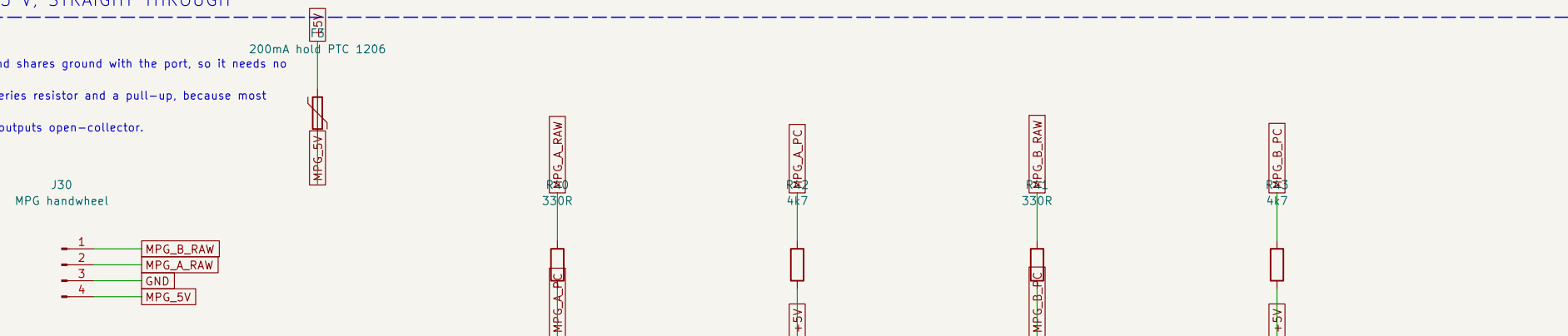
4k7 in 1206: $(24-1.2)^2/4700 = 0.11 \text{ W}$, which a 0603 cannot hold.



HANDWHEEL – 5 V, STRAIGHT THROUGH

The MPG runs at 5 V and shares ground with the port, so it needs no

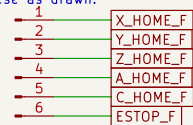
handwheels drive their outputs open-collector.



EMG & HOME – ONE POSITION PER SIGNAL

One six-way block, labelled EMG C A Z Y X as on the board in service.

The sensors take their +24 and 0 V from the 24 V terminal, not from a pin each. PNP (sourcing) sensors: an NPN will not drive these as drawn.



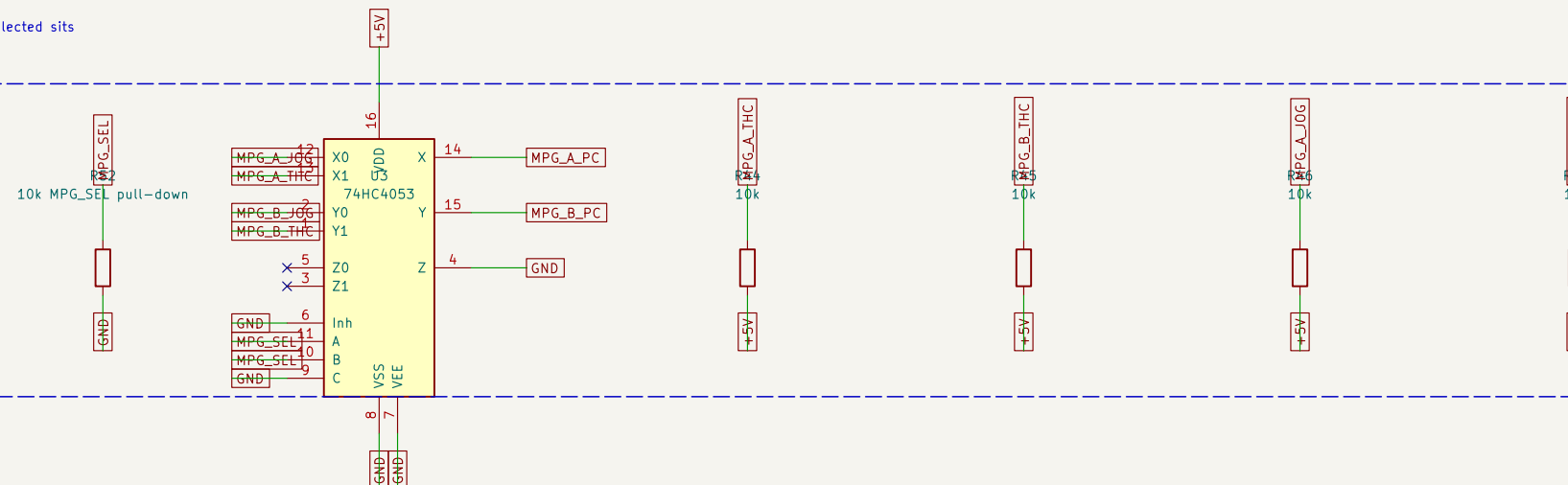
HANDWHEEL ROUTING – JOG OR THC, CHOSEN BY SendSerial

One encoder, two jobs. U3 sends MPG A and B either to port 2 pins 12

and 13, where Mach3 reads them as the handwheel, or to pins 10 and 11,

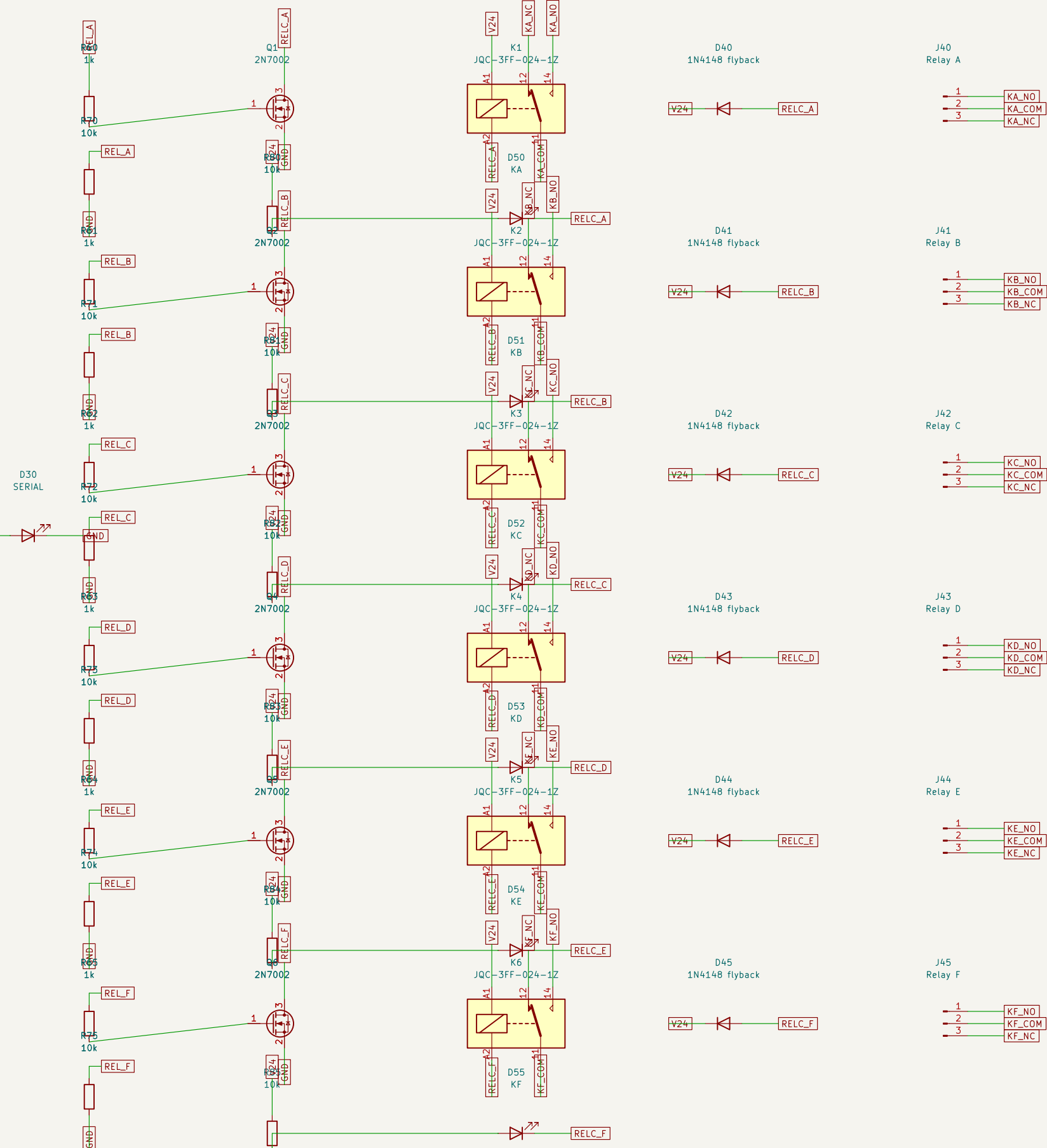
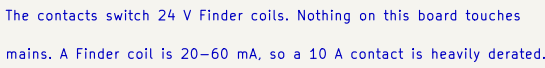
where it reads them as THC up and down. The MCU picks, on SendSerial

The four LPT pins are pulled up, so whichever pair is not selected sits



protocol as the board in service, so existing macros keep working.

SIX RELAYS – 24 V COILS ONLY



Title: RS232, MCU and six relays

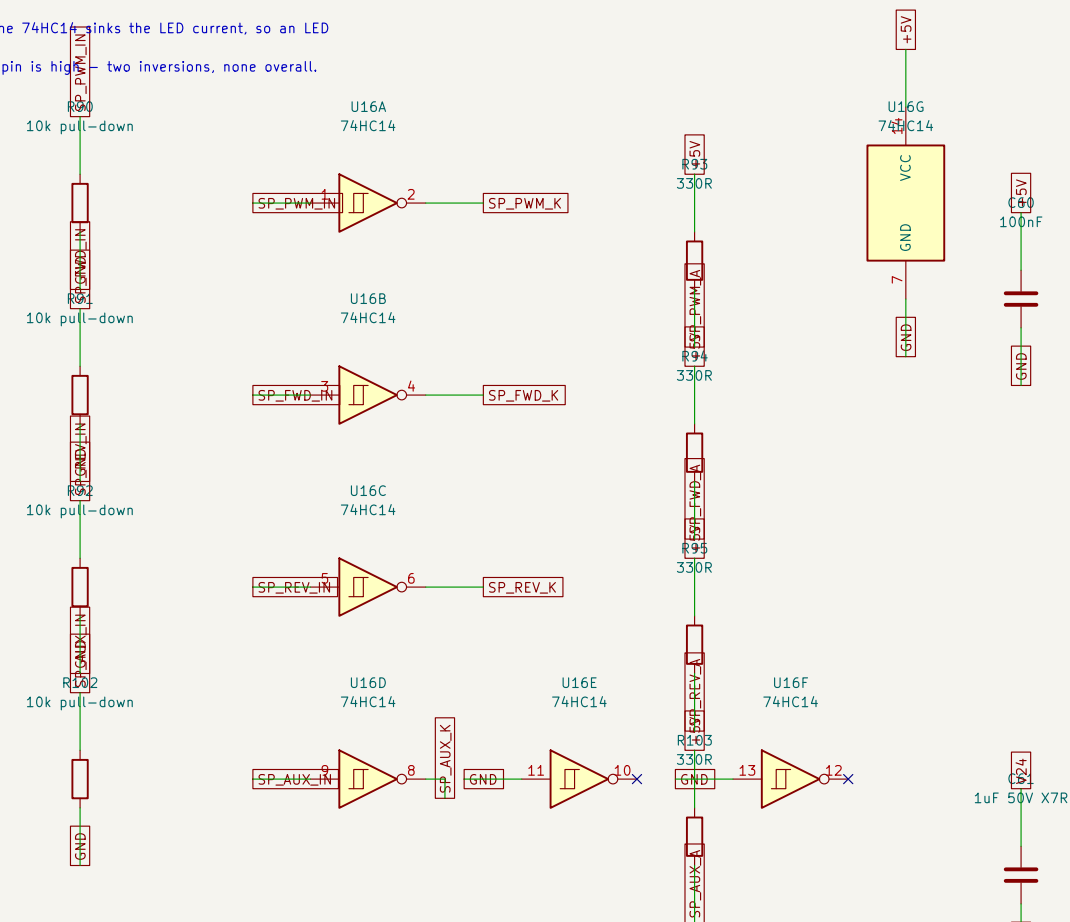
ate:

Rev: A1 ENGINEERING ONLY

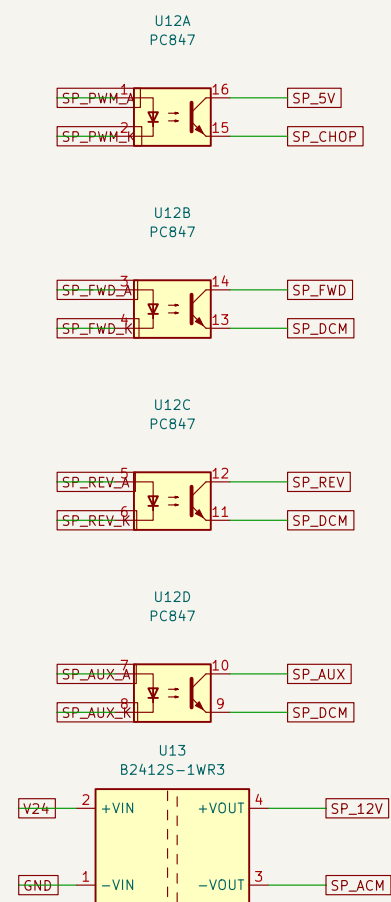
Id: 5/6

Mach3: port 2 pin 7 = spindle PWM, PWMBase Freq 100 Hz. Pin 8 = Output #1 (M3, FWD).
pin 9 = Output #2 (M4, REV). All three active high. Trim RV1 for 10.0 V at full speed.

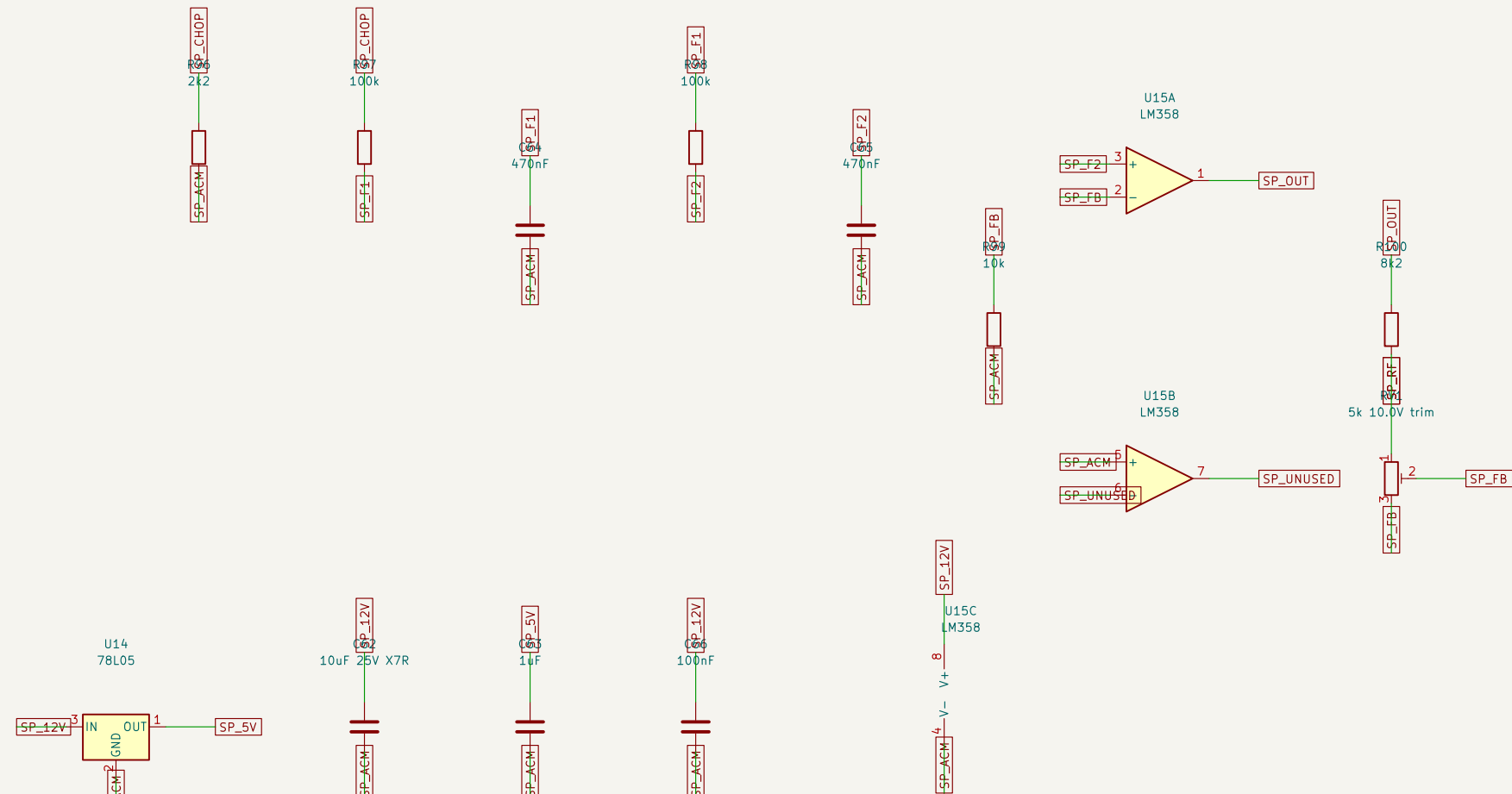
Pulled down, so with the PC off or the cable out every LED is dark:
no speed, no direction. The 74HC14 sinks the LED current, so an LED
lights only when its port pin is high – two inversions, none overall.



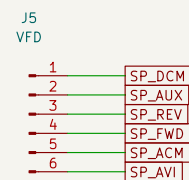
Nothing crosses this box but light
and the DC-DC module.



SP_ACM and SP_DCM are the VFD commons. Neither touches GND.



AVI ACM FWD REV AUX DCM. Most VFDs ship with FWD/REV in NPN (sink) mode, which is what these outputs need. PC847: 35 V, 50 mA per output.



TP4 ACM	TP5 12V	TP6 12V
SP_ACM	SP_12V	SP_12V